

Labwork 2: Get to Know Your GPU

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1 Objective

Use Numba CUDA to obtain the GPU name, multiprocessor count, CUDA core count, and memory size.

2 Implementation

The program selects GPU 0 using `cuda.select_device(0)` and reads its name and multiprocessor count. The CUDA core count is calculated as:

$$\text{CUDA cores} = \text{SM count} \times \text{cores per SM}$$

Memory size is read with `get_memory_info()` and converted to GiB.

3 Results

The program was run on WSL2 using the Conda environment m2.

```
Device name: NVIDIA GeForce RTX 5060 Ti
Multiprocessor count: 36
CUDA core count: 4608
Memory: 15.90 GiB
```

The GPU has 36 multiprocessors and 128 CUDA cores per multiprocessor, giving $36 \times 128 = 4608$ CUDA cores. The memory reported by the CUDA context is approximately 15.90 GiB.